

The Goal

Let

$$f_d = \sum_{k=1}^c k^d.$$

We must compute f_d for all d from 0 to $n - 1$ in subquadratic time.

Solution

Let

$$F(x) = \sum_{d \geq 0} f_d \frac{x^d}{d!}$$

be the **exponential** generating function of f . Let's do some algebra:

$$\begin{aligned} F(x) &= \sum_{d \geq 0} \left(\sum_{k=1}^c k^d \right) \frac{x^d}{d!} && \text{substitute in the definition of } f_d \\ &= \sum_{k=1}^c \sum_{d \geq 0} k^d \frac{x^d}{d!} && \text{interchange order of summations} \\ &= \sum_{k=1}^c \sum_{d \geq 0} \frac{(kx)^d}{d!} && \text{push } k^d \text{ and } x^d \text{ in together} \\ &= \sum_{k=1}^c e^{kx} && \text{that's now the Taylor series of } e^{kx} \\ &= \sum_{k=1}^c (e^x)^k && \text{use laws of exponents to write it like this, so that...} \\ &= e^x \frac{e^{cx} - 1}{e^x - 1} && \text{it's a geometric series.} \\ &= e^x \frac{e^{cx} - 1}{x} \left(\frac{e^x - 1}{x} \right)^{-1} && \text{rewrite it like this to make the denominator invertible} \end{aligned}$$

Note that we used the exponential generating function of f , and *not* just the ordinary power series generating function, specifically so that the summation could be compressed into a geometric series.

In summary,

$$F(x) = e^x \frac{e^{cx} - 1}{x} \left(\frac{e^x - 1}{x} \right)^{-1}$$

is the exponential generating function of f .

If we can compute the first n terms of this power series, then we have f_d for all d from 0 to $n - 1$.

- To find the first n terms in the generating function of e^x ...
 - The Taylor series of e^x is standard.
- To find the first n terms in the generating function of $\frac{e^{kx} - 1}{x}$...
 - The Taylor series of e^{kx} is standard.
 - Then, just remove the constant term then shift the coefficients over by 1
- To find the first n terms in the generating function of $\left(\frac{e^x - 1}{x}\right)^{-1}$...
 - The Taylor series of e^x is standard.
 - Then, just remove the constant term then shift the coefficients over by 1
 - Then, you can computationally find the power series of the reciprocal in $O(n \lg n)$ by some "standard" tricks (for some definition of standard).

Multiply these three together (using FFT) and we will have computed all the terms we want in $O(n \lg n)$.